CHEBOTAREV DENSITY THEOREM AND ASYMPTOTIC DENSITY OF RANK-METRIC CODES

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Abstract. We study the asymptotic density of pointless sections on geometrically irreducible varieties over finite fields, and apply the results to rank-metric codes in the setting of determinantal varieties. This approach recovers known results in the sparse and abundant cases with concise proofs, and also settles the threshold case. To address the latter, we introduce the notion of quasireflexive varieties, show that determinantal varieties satisfy this property, and then invoke the Chebotarev density theorem for varieties over finite fields to obtain the desired estimate.

1. Introduction

Let $\mathbb{F}_q$ be the finite field with q elements, where q is a prime power, and regard $\mathbb{F}_q^{mn}$ as the space of $m \times n$ matrices over $\mathbb{F}_q$ equipped with the rank metric

$$d(A, B) := \operatorname{rank}(A - B), \qquad A, B \in \mathbb{F}_q^{mn}.$$

Throughout the paper, we assume without loss of generality that

$$m \geq n.$$

A *rank-metric code* is a linear subspace $V \subseteq \mathbb{F}_q^{mn}$. Its *rank distance*, which measures the code's error-correcting capability, is defined by

$$\begin{aligned} d_R(V) &:= \min \{ d(A, B) \mid A, B \in V,\ A \neq B \} \\ &= \min \{ \operatorname{rank}(M) \mid M \in V \setminus \{0\} \}. \end{aligned}$$

Rank-metric codes have become a central object of study in modern coding theory. They arise natural in settings where errors are measured by rank, with important applications to random network coding and secure network coding [SKK08, SK11]. They are also closely connected to areas of pure mathematics, including q-polymatroids and finite geometry [GJLR20, CMPZ17]. For a general introduction to the subject, we refer the reader to [BHL+22].

One natural problem in this area is to estimate the density of such codes with prescribed parameters. Fix an integer δ and consider the density of k-dimensional codes with rank distance at least δ:

$$D_q(m \times n, \delta, k) := \frac{\left|\{\text{codes } V \subseteq \mathbb{F}_q^{mn} \mid d_R(V) \geq \delta,\ \dim V = k\}\right|}{\left|\{\text{codes } V \subseteq \mathbb{F}_q^{mn} \mid \dim V = k\}\right|}.$$

By Gruica and Ravagnani [GR22, Theorem 5.12], one has

$$\lim_{q \to \infty} D_q(m \times n, \delta, k) = \begin{cases} 0 & \text{if } \ (\delta - 1)(m + n - \delta + 1) > mn - k + 1, \\ 1 & \text{if } \ (\delta - 1)(m + n - \delta + 1) < mn - k + 1. \end{cases}$$

In view of this result, we refer to the first situation as the *sparse case* and to the second as the *abundant case*.

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The remaining situation is the *threshold case*, namely when

$$(\delta-1)(m+n-\delta+1)=mn-k+1. \tag{1.1}$$

In this case, the limit has been computed in the specific setting where the codes attain the *Singleton bound*. This bound asserts that the dimension k and the rank distance δ of a rank-metric code satisfy

$$k\le m(n-\delta+1).$$

Codes achieving equality in this bound are called *maximum rank-distance (MRD) codes*. For such codes, substituting $k=m(n-\delta+1)$ into (1.1) yields $n=\delta=2$. In this special case, both [AGL19, Proposition VII.5] and [GR22, Theorem 5.9] show that

$$\lim_{q\to\infty} D_q(m\times 2,2,m)=\sum_{i=0}^{m}\frac{(-1)^i}{i!}.$$

Another known result, also due to Gruica and Ravagnani [GR22, Proposition 4.4], states for the threshold case that

$$\limsup_{q\to\infty} D_q(m\times n,\delta,m(n-\delta+1))\le\frac{1}{2}.$$

In this paper, we analyze the density function within the framework of determinantal varieties. In the projective space $\mathbb{P}(\mathbb{F}_q^{mn})\cong\mathbb{P}^{mn-1}$, matrices of ranks less than a given integer δ, where $2\le\delta\le n$, form an algebraic variety

$$Y_\delta:=\left\{[M]\in\mathbb{P}^{mn-1}\mid \operatorname{rank}(M)<\delta\right\}.$$

By definition, a code $V\subseteq\mathbb{F}_q^{mn}$ has $d_R(V)\ge\delta$ if and only if the linear section $\mathbb{P}(V)\cap Y_\delta$ contains no $\mathbb{F}_q$-point. Moreover, since

$$\dim Y_\delta=(mn-1)-(m-\delta+1)(n-\delta+1),$$

Bertini's theorem asserts that for a general linear subspace $V\subseteq\overline{\mathbb{F}}_q^{mn}$ of dimension k,

$$\mathbb{P}(V)\cap Y_\delta \text{ is } \begin{cases}\text{positive-dimensional} & \text{if } (\delta-1)(m+n-\delta+1)>mn-k+1,\\ \text{empty} & \text{if } (\delta-1)(m+n-\delta+1)<mn-k+1,\\ \text{zero-dimensional} & \text{if } (\delta-1)(m+n-\delta+1)=mn-k+1,\end{cases}$$

which correspond precisely to the sparse, abundant, and threshold cases, respectively.

Determinantal varieties are examples of *geometrically irreducible varieties*, meaning that they remain irreducible in the Zariski topology after extending the base field to the algebraic closure $\overline{\mathbb{F}}_q$; see, for example, [Har92, Proposition 12.2]. They also satisfy *quasireflexivity*, a notion originally introduced for curves by Entin [Ent21, Definition at p. 8]. More precisely, a curve $C\subseteq\mathbb{P}^n$ of degree d is *quasireflexive* if there exists a hyperplane whose intersection with C consists of $d-1$ points. Thus the tangency is as simple as possible, while all other intersections are transverse. We extend this notion to higher dimensions as follows.

Definition 1.1. A projective variety $Y\subseteq\mathbb{P}^N$ of degree Δ is *quasireflexive* if there exists a linear subspace $L\subseteq\mathbb{P}^N$ of dimension $N-\dim Y$ whose intersection with Y consists of $\Delta-1$ points at which Y is smooth.

Quasireflexivity enables us to apply the *Chebotarev density theorem*; see Section 3 for further details. The main result in this paper is as follows.

Theorem 1.2. *Let $Y \subseteq \mathbb{P}^N$ be a geometrically irreducible variety over $\mathbb{F}_q$ of degree Δ, and let $\mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)$ be the Grassmannian of ℓ-dimensional linear subspaces in $\mathbb{P}^N$. Consider the density function*

$$D_q(Y,\ell) = \frac{\left|\left\{L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \;\middle|\; (L \cap Y)(\mathbb{F}_q) = \emptyset\right\}\right|}{|\mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q)|}.$$

(1) *There exists a constant c_1 depending only on N, Δ, ℓ such that*

$$0 \leq D_q(Y,\ell) \leq c_1 q^{N-\dim Y-\ell}.$$

In particular, the condition $\dim Y + \ell > N$ implies

$$\lim_{q\to\infty} D_q(Y,\ell) = 0.$$

(2) *There exists a constant c_2 depending only on N, Δ, ℓ such that*

$$1 - c_2 q^{\dim Y+\ell-N} \leq D_q(Y,\ell) \leq 1.$$

In particular, the condition $\dim Y + \ell < N$ implies

$$\lim_{q\to\infty} D_q(Y,\ell) = 1.$$

(3) *Suppose $\dim Y + \ell = N$ and Y is quasireflexive. Then*

$$\lim_{q\to\infty} D_q(Y,\ell) = \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}.$$

We prove that determinantal varieties are quasireflexive in Theorem 4.12. To obtain the asymptotic density of rank-metric codes, we apply Theorem 1.2 to the determinantal variety

$$Y = Y_\delta \subseteq \mathbb{P}^{mn-1}$$

with

$$\ell = k-1 \qquad \text{and} \qquad \Delta = \deg(Y_\delta) = \prod_{i=0}^{n-\delta} \frac{\binom{m+i}{\delta-1}}{\binom{\delta-1+i}{\delta-1}}.$$

In this setting, we have $D_q(m\times n, \delta, k) = D_q(Y_\delta,\, k-1)$.

Corollary 1.3. *The asymptotic density of k-dimensional codes in $\mathbb{F}_q^{mn}$ with rank distance at least δ as $q \to \infty$ is given by*

$$\lim_{q\to\infty} D_q(m\times n,\delta,k) = \begin{cases} 0 & \textit{if} \quad (\delta-1)(m+n-\delta+1) > mn-k+1, \\ 1 & \textit{if} \quad (\delta-1)(m+n-\delta+1) < mn-k+1, \\ \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!} & \textit{if} \quad (\delta-1)(m+n-\delta+1) = mn-k+1 \end{cases}$$

Organization of the paper. The first two cases of Theorem 1.2 are proved in Section 2, while the remaining threshold case is treated in Section 3. In Section 4, we verify that determinantal varieties are quasireflexive. In Appendix A, we discuss the closely related notion of reflexivity, and prove that determinantal varieties satisfy this property in characteristic different from 2.

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2. Asymptotic density in the sparse and abundant cases

We prove the sparse and abundant cases of Theorem 1.2 in this section.

2.1. **The sparse case.** For any variety $Y \subseteq \mathbb{P}^N$, the assignment

$$\begin{aligned} \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) &\longrightarrow \mathbb{Z}_{\geq 0} \\ L &\longmapsto |(L \cap Y)(\mathbb{F}_q)| \end{aligned}$$

defines a random variable. As shown in [PS22, Lemma 4.1] for a special case and in [KS22, Lemma 2.1] more generally, the mean μ and variance σ^2 of this random variable satisfy

$$\mu = |Y(\mathbb{F}_q)| \left(q^{\ell-N} + O\left(q^{\ell-N-1}\right)\right), \quad \sigma^2 = O\left(q^{\ell-N}|Y(\mathbb{F}_q)|\right).$$

If Y is geometrically irreducible, the Lang–Weil bound [LW54, Theorem 1] gives

$$|Y(\mathbb{F}_q)| = q^{\dim Y} + O\left(q^{\dim Y - \frac{1}{2}}\right).$$

Substituting this into the previous expressions, we obtain

$$\mu = q^{\dim Y + \ell - N} + O\left(q^{\dim Y + \ell - N - \frac{1}{2}}\right), \quad \sigma^2 = q^{\dim Y + \ell - N} + O\left(q^{\dim Y + \ell - N - \frac{1}{2}}\right) \tag{2.1}$$

Proof of Theorem 1.2 (1). The inequality $D_q(Y, \ell) \geq 0$ holds by definition, so it remains to show that there exists a constant c_1, depending only on N, Δ, ℓ, such that

$$D_q(Y, \ell) \leq c_1 q^{N - \dim Y - \ell}.$$

Consider the set of pointless linear sections

$$\mathcal{U}_q := \left\{ L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \;\middle|\; (L \cap Y)(\mathbb{F}_q) = \emptyset \right\}.$$

For any $L \in \mathcal{U}_q$, we have $|(L \cap Y)(\mathbb{F}_q)| = 0$. Hence, for any $\epsilon < 1$,

$$||(L \cap Y)(\mathbb{F}_q)| - \mu| = \mu \geq \epsilon\mu = \left(\frac{\epsilon\mu}{\sigma}\right)\sigma.$$

It follows that

$$\begin{aligned} \mathrm{Prob}\left(L \in \mathcal{U}_q\right) &\leq \mathrm{Prob}\left(||(L \cap Y)(\mathbb{F}_q)| - \mu| \geq \left(\frac{\epsilon\mu}{\sigma}\right)\sigma\right) \\ \text{(Chebyshev's inequality)} \qquad &\leq \left(\frac{\sigma}{\epsilon\mu}\right)^2 \\ \text{(Equation (2.1))} \qquad &= \frac{1}{\epsilon^2} O\left(q^{N-\dim Y - \ell}\right) = O\left(q^{N - \dim Y - \ell}\right). \end{aligned}$$

This establishes the desired upper bound. In particular, the condition $\dim Y + \ell > N$ implies $\lim_{q\to\infty} q^{N-\dim Y - \ell} = 0$, which further implies $\lim_{q\to\infty} D_q(Y, \ell) = 0$. $\square$

Remark 2.1. Our proof for the limit $\lim_{q\to\infty} D_q(Y, \ell) = 0$ shows more precisely that

$$D_q(Y, \ell) = O(q^{N - \dim Y - \ell}).$$

However, the exponent $N - \dim Y - \ell$ (the rate of convergence) is not necessarily optimal. For instance, when $n = m = k = \delta = 3$, our upper bound gives $O(q^{-1})$, whereas Gluesing-Luerssen obtained a sharper bound of $O(q^{-3})$ in this particular case using semifield theory [GL20, Theorem 2.4].

2.2. **The abundant case.** Consider a geometrically irreducible variety $Y \subseteq \mathbb{P}^N$ defined over $\mathbb{F}_q$, together with the incidence correspondence

$$\mathcal{X} := \left\{(y, L) \in Y \times \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) \;\middle|\; y \in L\right\}.$$

There are two natural projection maps

$$\begin{tikzcd} & \mathcal{X} \arrow[dl, "\pi_Y"'] \arrow[dr, "\pi_{\mathrm{Gr}}"] & \\ Y & & \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N). \end{tikzcd}$$

where the image of the second projection is

$$\mathcal{N} := \pi_{\mathrm{Gr}}(\mathcal{X}) = \left\{L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) \;\middle|\; L \cap Y \neq \emptyset\right\}.$$

Lemma 2.2. *The incidence correspondence $\mathcal{X}$ is geometrically irreducible with*

$$\dim \mathcal{X} = \dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) - (N - \dim Y - \ell).$$

Proof. For any $y \in Y$, the fiber $\pi_Y^{-1}(y)$ consists of ℓ-dimensional linear subspaces of $\mathbb{P}^N$ containing y. Projecting from y identifies such subspaces with $(\ell - 1)$-dimensional linear subspaces of $\mathbb{P}^{N-1}$, yielding an isomorphism

$$\pi_Y^{-1}(y) \cong \mathrm{Gr}(\mathbb{P}^{\ell-1}, \mathbb{P}^{N-1}).$$

Hence, the fibers of π_Y are geometrically irreducible and have the same dimension. Since Y is geometrically irreducible, it follows that $\mathcal{X}$ is geometrically irreducible by [Sha13, Theorem 1.26].

For the dimension, we compute

$$\begin{aligned} \dim \mathcal{X} = \dim \pi_Y^{-1}(y) + \dim Y &= \dim \mathrm{Gr}(\mathbb{P}^{\ell-1}, \mathbb{P}^{N-1}) + \dim Y \\ &= \ell(N-\ell) + \dim Y \\ &= \ell(N-\ell) + (N-\ell) + \dim Y - (N-\ell) \\ &= (\ell+1)(N-\ell) + \dim Y - (N-\ell) \\ &= \dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) - (N - \dim Y - \ell). \end{aligned}$$

This proves the statement. ☐

Lemma 2.3. *The projection image $\mathcal{N} = \pi_{\mathrm{Gr}}(\mathcal{X})$ is geometrically irreducible with*

$$\dim \mathcal{N} \leq \dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) - (N - \dim Y - \ell).$$

Proof. The variety $\mathcal{N}$ is geometrically irreducible since $\mathcal{X}$ is geometrically irreducible by Lemma 2.2. The upper bound on the dimension follows from the inequality $\dim \mathcal{N} \leq \dim \mathcal{X}$ together with Lemma 2.2. ☐

Proof of Theorem 1.2 (2). By definition,

$$1 - D_q(Y, \ell) = \frac{|\mathcal{N}(\mathbb{F}_q)|}{|\mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q)|}. \tag{2.2}$$

The variety $\mathcal{N}$ is geometrically irreducible by Lemma 2.3, so we can apply the Lang–Weil bound [LW54, Theorem 1] to obtain

$$\left||\mathcal{N}(\mathbb{F}_q)| - q^{\dim \mathcal{N}}\right| \leq A q^{\dim \mathcal{N} - 1/2}$$

where A is a constant depending only on N, Δ, ℓ. From this inequality, one can check that for $q \geq \max\{A^2, 1\}$,

$$|\mathcal{N}(\mathbb{F}_q)| \leq q^{\dim \mathcal{N}} + Aq^{\dim \mathcal{N}-1/2} \leq 2q^{\dim \mathcal{N}}.$$

Substituting this into (2.2) and using Lemma 2.3, we get for $q \geq \max\{A^2, 1\}$,

$$1 - D_q(Y, \ell) \leq \frac{2q^{\dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)-(N-\dim Y-\ell)}}{q^{\dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)}} = 2q^{\dim Y+\ell-N}.$$

Hence for all q,

$$1 - D_q(Y, \ell) \leq c_2 q^{\dim Y+\ell-N}$$

for some constant c_2 depending only on N, Δ, ℓ. Rearranging the terms gives

$$1 - c_2 q^{\dim Y+\ell-N} \leq D_q(Y, \ell) \leq 1.$$

If $\dim Y + \ell < N$, then $\lim_{q\to\infty}(1 - c_2 q^{\dim Y+\ell-N}) = 1$, and thus $\lim_{q\to\infty} D_q(Y, \ell) = 1$. $\square$

3. Asymptotic density in the threshold case

Consider a geometrically irreducible variety $Y \subseteq \mathbb{P}^N$ of degree Δ over $\mathbb{F}_q$, together with the incidence correspondence

$$\begin{tikzcd}[column sep=small] & \mathcal{X} := \left\{(y, L) \in Y \times \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) \,\middle|\, y \in L\right\} \arrow[dl, "\pi_Y"'] \arrow[dr, "\pi_{\mathrm{Gr}}"] & \\ Y & & \mathcal{G} := \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N). \end{tikzcd}$$

Assume that $\dim Y + \ell = N$. Then a general ℓ-plane $L \in \mathcal{G}$ intersects Y transversely in Δ points by Bertini's theorem. In particular, the map

$$\pi_{\mathrm{Gr}} \colon \mathcal{X} \longrightarrow \mathcal{G}$$

is finite and generically étale with $\deg(\pi_{\mathrm{Gr}}) = \Delta$. In this setting, Theorem 1.2 (3) concerns the distribution of $\mathbb{F}_q$-points in $\mathcal{G}$ whose preimage under π_{Gr} has no $\mathbb{F}_q$-points.

3.1. **Chebotarev density theorem.** Before proceeding to the proof, we briefly review the background for the key ingredient. The main source for the following material is [Ent21, Sections 3 and 4]. Let $\mathcal{G}^\circ$ be a geometrically irreducible quasi-projective variety, and let

$$f \colon \mathcal{X}^\circ \longrightarrow \mathcal{G}^\circ$$

be a finite étale morphism of degree Δ. For a geometric point $L_0 \in \mathcal{G}^\circ$, the étale fundamental group $\pi^{\mathrm{et}}(\mathcal{G}^\circ, L_0)$ acts on the fiber $f^{-1}(L_0)$, giving rise to a homomorphism to the symmetric group S_Δ. The *arithmetic monodromy group* $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ is defined as the image of this homomorphism. This group is well defined up to conjugation since different choices of the base point L_0 yield conjugate subgroups of S_Δ.

On the other hand, the *geometric monodromy group* is defined by

$$\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) := \mathrm{Mon}\left(\mathcal{X}^\circ \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}/\mathcal{G}^\circ \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}\right).$$

It is related to $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ via the short exact sequence

$$1 \longrightarrow \mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) \longrightarrow \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \longrightarrow \mathrm{Gal}(\mathbb{F}_{q^\nu}/\mathbb{F}_q) \longrightarrow 1,$$

where ν is the minimal positive integer such that

$$\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) = \mathrm{Mon}\left(\mathcal{X}^\circ \times_{\mathbb{F}_q} \mathbb{F}_{q^\nu}/\mathcal{G}^\circ \times_{\mathbb{F}_q} \mathbb{F}_{q^\nu}\right).$$

In particular, the arithmetic and geometric monodromy groups coincide when q is large enough.

Over each point $L \in \mathcal{G}^\circ(\mathbb{F}_q)$, the fiber $f^{-1}(L) \subseteq \mathcal{X}^\circ$ is a 0-dimensional scheme over $\mathbb{F}_q$ consisting of Δ geometric points. The action of the Frobenius endomorphism on this fiber determines a conjugacy class

$$\mathrm{Frob}_q(L) \subseteq \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ).$$

Note that $f^{-1}(L)$ contains no $\mathbb{F}_q$-points if and only if the class $\mathrm{Frob}_q(L)$ consists of derangements. Thus our problem reduces to estimating the density of points $L \in \mathcal{G}^\circ(\mathbb{F}_q)$ for which $\mathrm{Frob}_q(L)$ consists of derangements.

The key tool for our argument is the following version of Chebotarev density theorem proved by Entin [Ent21]. We state only the form relevant to our setting, namely the case in which the arithmetic and geometric monodromy groups coincide.

Theorem 3.1 ([Ent21, Theorem 3])**.** *Let $f\colon \mathcal{X}^\circ \to \mathcal{G}^\circ$ be a finite étale morphism between quasiprojective varieties over $\mathbb{F}_q$ with $\mathcal{G}^\circ$ geometrically irreducible. Suppose that q is sufficiently large so that*

$$\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) = \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$$

Then for each conjugacy class $\mathcal{C} \subseteq \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$, we have

$$|\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \mid \mathrm{Frob}_q(L) = \mathcal{C}\}| = \frac{|\mathcal{C}|}{|\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)|} q^{\dim \mathcal{G}^\circ} \left(1 + O\left(q^{-1/2}\right)\right).$$

3.2. **Proof for the threshold case.** We now retain the notation introduced in the beginning of this section. In our application, $\mathcal{G}^\circ \subseteq \mathcal{G}$ is a Zariski open subset over which π_{Gr} is étale, and we set $\mathcal{X}^\circ := \pi_{\mathrm{Gr}}^{-1}(\mathcal{G}^\circ)$. The restriction

$$f := \pi_{\mathrm{Gr}}|_{\mathcal{X}^\circ} \colon \mathcal{X}^\circ \longrightarrow \mathcal{G}^\circ$$

is therefore a finite étale morphism of degree Δ.

Lemma 3.2. *If the variety Y is quasireflexive, then* $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \cong S_\Delta$.

Proof. We fix a point $L_0 \in \mathcal{G}^\circ(\mathbb{F}_q)$ and identify $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ as a subgroup of S_Δ by considering its action on the fiber $f^{-1}(L_0)$. Let

$$D \subseteq \mathcal{X} \times_{\mathcal{G}} \mathcal{X} \qquad \text{and} \qquad D' \subseteq Y \times Y$$

denote the diagonals. The fiber of the morphism

$$\mathcal{X} \times_{\mathcal{G}} \mathcal{X} \setminus D \longrightarrow Y \times Y \setminus D'$$

over $(y_1, y_2) \in Y \times Y \setminus D'$ consists of the points $((y_1, L), (y_2, L))$, where $L \in \mathcal{G} = \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)$ contains the line spanned by y_1 and y_2. Thus $\mathcal{X} \times_{\mathcal{G}} \mathcal{X} \setminus D$ has the structure of a Grassmannian bundle over the irreducible variety $Y \times Y \setminus D'$, and is therefore itself irreducible. It follows that $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ acts doubly transitively on $f^{-1}(L_0)$ by [BH86, Proposition 2 (ii)].

It remains to show that $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ contains a simple transposition. Since Y is quasireflexive, there exists $L \in \mathcal{G}$ meeting the smooth locus of Y in $\Delta - 1$ points $y_0, y_1, \ldots, y_{\Delta-2}$, where y_0 has multiplicity two and all other intersections are transverse. The map π_{Gr} is étale at $(y_i, L) \in \mathcal{X}$ for all $i = 1, \ldots, \Delta - 2$. Moreover, $\mathcal{X}$ is locally irreducible at (y_0, L), since it is a Grassmannian bundle over Y and Y is smooth at y_0. Therefore, $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ contains a simple transposition by [BH86, Proposition 3], and hence is isomorphic to S_Δ. $\square$

Lemma 3.3. *If the variety Y is quasireflexive, then for q sufficiently large,*

$$\left|\left\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \;\middle|\; \mathrm{Frob}_q \text{ deranges } \pi_{\mathrm{Gr}}^{-1}(L)\right\}\right| = \left(\sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}\right) q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right).$$

Proof. By Lemma 3.2, we have $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \cong S_\Delta$. Applying Theorem 3.1, we obtain

$$\begin{aligned}
\left|\left\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \;\middle|\; \mathrm{Frob}_q \text{ deranges } \pi_{\mathrm{Gr}}^{-1}(L)\right\}\right| &= \sum_{\substack{\mathcal{C} \text{ consists of} \\ \text{derangements}}} \frac{|\mathcal{C}|}{|S_\Delta|} q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right) \\
&= \frac{|\{\text{derangements in } S_\Delta\}|}{|S_\Delta|} q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right) \\
&= \left(\sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}\right) q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right),
\end{aligned}$$

as claimed. □

Proof of Theorem 1.2 (3). By definition,

$$D_q(Y, \ell) = \frac{|\{L \in \mathcal{G}(\mathbb{F}_q) \mid (L \cap Y)(\mathbb{F}_q) = \emptyset\}|}{|\mathcal{G}(\mathbb{F}_q)|}.$$

The condition $(L \cap Y)(\mathbb{F}_q) = \emptyset$ holds if and only if the Frobenius endomorphism acts without fixed points on the fiber $\pi_{\mathrm{Gr}}^{-1}(L)$. Hence, over the open subset $\mathcal{G}^\circ \subseteq \mathcal{G}$, Lemma 3.3 gives

$$|\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \mid (L \cap Y)(\mathbb{F}_q) = \emptyset\}| = \left(\sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}\right) q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right).$$

It remains to estimate the contribution from the complement $\mathcal{D} := \mathcal{G} \setminus \mathcal{G}^\circ$. Since $\mathcal{D}$ is a proper closed subset, we have $\dim \mathcal{D} < \dim \mathcal{G}$. By the Lang–Weil bound [LW54, Theorem 1],

$$|\mathcal{D}(\mathbb{F}_q)| \leq O\left(q^{\dim \mathcal{G} - 1}\right), \qquad |\mathcal{G}(\mathbb{F}_q)| = q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right).$$

Comparing these two terms shows that the contribution from $\mathcal{D}(\mathbb{F}_q)$ is negligible. Therefore,

$$D_q(Y, \ell) = \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!} + O\left(q^{-1/2}\right).$$

In particular,

$$\lim_{q \to \infty} D_q(Y, \ell) = \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}.$$

This completes the proof. □

4. Quasireflexivity of determinantal varieties

In this section, we prove that determinantal varieties are quasireflexive over fields of arbitrary characteristic. Throughout, we fix the following notation. Over an algebraically closed field K, we consider the determinantal variety

$$Y_\delta \subseteq \mathbb{P}^N \qquad \text{where} \qquad N = mn - 1$$

consisting of $m \times n$ matrices of rank $< \delta$, and assume without loss of generality that

$$m \geq n.$$

For brevity, we denote its dimension and codimension by

$$\begin{aligned} d &:= \dim Y_\delta = N - (m-\delta+1)(n-\delta+1), \\ c &:= \operatorname{codim} Y_\delta = (m-\delta+1)(n-\delta+1). \end{aligned}$$

By definition, to show that Y_δ is quasireflexive, we need to find a c-dimensional linear subspace $L \subseteq \mathbb{P}^N$ that intersects Y_δ in $\deg(Y_\delta) - 1$ points all lying in the smooth locus

$$Y_\delta^{\mathrm{sm}} \subseteq Y_\delta.$$

We begin by treating the hypersurface case.

Proposition 4.1. *If Y_δ is a hypersurface, that is, if $m = n = \delta$, then it is quasireflexive.*

Proof. In this setting, $Y_\delta \subseteq \mathbb{P}^N = \mathbb{P}^{n^2-1}$ is a hypersurface of degree n. It therefore suffices to construct a line $L \subseteq \mathbb{P}^N$ meeting the smooth locus Y_δ^{sm} in $n-1$ distinct points.

Choose $n-1$ distinct points $[a_i : b_i] \in \mathbb{P}^1$, where $i = 1, \ldots, n-1$, and consider the morphism

$$\varphi \colon \mathbb{P}^1 \longrightarrow \mathbb{P}^N, \quad \varphi(s:t) = \begin{pmatrix} a_1 s + b_1 t & a_2 s + b_2 t & 0 & \cdots & 0 \\ 0 & a_1 s + b_1 t & 0 & \cdots & 0 \\ 0 & 0 & a_2 s + b_2 t & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{n-1} s + b_{n-1} t \end{pmatrix}.$$

When $n = 2$, we interpret this map as

$$\varphi \colon \mathbb{P}^1 \longrightarrow \mathbb{P}^3 \quad \varphi(s:t) = \begin{pmatrix} a_1 s + b_1 t & a_2 s + b_2 t \\ 0 & a_1 s + b_1 t \end{pmatrix}.$$

Let $L := \varphi(\mathbb{P}^1)$, which is a line in $\mathbb{P}^N$. The scheme-theoretic intersection $L \cap Y_\delta$ is defined by the determinant of $\varphi(s:t)$, hence by the polynomial equation

$$(a_1 s + b_1 t)^2 (a_2 s + b_2 t) \cdots (a_{n-1} s + b_{n-1} t) = 0.$$

This equation has a double root at $[b_1 : -a_1]$ and $n-2$ simple roots at $[b_i : -a_i]$ for $i \geq 2$.

For the matrix $\varphi(b_1 : -a_1)$, the first two diagonal entries vanish, while the $(1,2)$-entry equals $a_2 b_1 - a_1 b_2 \neq 0$, and all remaining diagonal entries are nonzero. Hence the matrix has rank $n-1$, so it represents a point in the smooth locus Y_δ^{sm}. Since $L \cap Y_\delta$ has degree n, and one intersection point is a smooth point of multiplicity two, the remaining intersection points must also lie in Y_δ^{sm}. Therefore, L meets Y_δ^{sm} in $n-1$ distinct points. This proves that Y_δ is quasireflexive. □

4.1. **Preliminary lemmas on tangent spaces.** The remainder of this section is devoted to the non-hypersurface case. Here, we collect properties of the tangent spaces of a determinantal variety that are needed for our proof.

For any point $P \in Y_\delta^{\mathrm{sm}}$, we write its embedded tangent space as $T_P Y_\delta \subseteq \mathbb{P}^N$. Recall that, as shown for example in [Har92, Example 14.16],

$$T_P Y_\delta = \left\{ Q \in \mathbb{P}^N \mid Q(\ker P) \subseteq \operatorname{im} P \right\}.$$

We will make use of the distinguished smooth point

$$P_0 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & 0 \end{pmatrix} \in Y_\delta^{\mathrm{sm}},$$

where $I_{\delta-1}$ denotes the $(\delta-1)\times(\delta-1)$ identity matrix. Its tangent space $T_{P_0}Y_\delta$ consists of points represented by matrices of the form

$$\begin{pmatrix} *_{\delta-1,\,\delta-1} & *_{\delta-1,\,n-\delta+1} \\ *_{m-\delta+1,\,\delta-1} & 0 \end{pmatrix} \tag{4.1}$$

where $*_{\alpha,\beta}$ denotes an arbitrary $\alpha \times \beta$ matrix.

We begin by analyzing the structure of the linear section $Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta$. This section admits a stratification according to the dimension of the intersection $\ker P_0 \cap \ker P \subseteq K^n$. More precisely, we have

$$Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta = \bigcup_{i=0}^{n-\delta+1} T_i$$

where

$$T_i = \{P \in Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta \mid \dim(\ker P_0 \cap \ker P) = (n-\delta+1) - i\}.$$

In particular, since $\ker P_0$ has dimension $n-\delta+1$, we have

$$T_0 = \{P \in Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta \mid \ker P_0 = \ker P\}.$$

Lemma 4.2. *For each $i \geq 0$, the Zariski closure $\overline{T_i} \subseteq Y_\delta^{\mathrm{sm}}$ is a cone with vertex P_0 and*

$$\begin{aligned} \dim T_i &= i(n-i) + (\delta-1-i)(m+i) - 1 \\ &= d - \left((\delta-1)(n-\delta+1-i) + i(m-n) + 2i^2\right). \end{aligned}$$

In particular, each T_i has positive codimension in Y_δ^{sm}. Moreover, only T_0 can have codimension one, and this occurs if and only if $\delta = n = 2$.

Proof. For a given $P \in T_i$, every point Q on the line spanned by P_0 and P is represented by a linear combination of P_0 and P. For a general such Q, its kernel satisfies the same dimension condition

$$\dim(\ker P_0 \cap \ker Q) = (n-\delta+1) - i.$$

Therefore, the line spanned by P_0 and P is contained in $\overline{T_i}$, showing that $\overline{T_i}$ is a cone with vertex P_0.

To compute the dimension of T_i, we first decompose

$$K^n \cong \ker P_0 \oplus K^{\delta-1}.$$

For each $P \in T_i$, the restrictions of a linear map $\widetilde{P}\colon K^n \to K^m$ representing it to the two summands behave as follows:

- Since $P \in T_{P_0}Y_\delta$, the restriction of $\widetilde{P}$ to $\ker P_0$ defines a linear map
  $$\ker P_0 \cong K^{n-\delta+1} \longrightarrow \operatorname{im} P_0 \cong K^{\delta-1}$$
  of rank i.
- The restriction of $\widetilde{P}$ to the second summand defines a linear map
  $$K^{\delta-1} \longrightarrow K^m$$
  of rank $\delta-1-i$.

Conversely, any linear map $K^n \to K^m$ whose restrictions to the two summands satisfy the above conditions determines a point of T_i. Hence, the dimension of the affine variety underlying T_i is given by the sum of the dimensions of the corresponding affine determinantal varieties, which yields

$$\dim T_i = i(n-i) + (\delta - 1 - i)(m+i) - 1.$$

To determine the codimension of T_i in Y_δ^{sm}, we compute

$$\begin{aligned}\dim Y_\delta - \dim T_i &= (\delta-1)(m+n-\delta+1) - (i(n-i) + (\delta-1-i)(m+i)) \\ &= (\delta-1)(n-\delta+1-i) + i(m-n) + 2i^2.\end{aligned}$$

This quantity is strictly positive since $m \geq n$ and

$$0 \leq i \leq n - \delta + 1,$$

while the equalities $i = 0$ and $i = n - \delta + 1$ cannot hold at the same time. Moreover, it is equal to 1 if and only if $i = 0$ and $\delta = n = 2$. This completes the proof. $\square$

Lemma 4.3. *For a general $P \in Y_\delta^{\mathrm{sm}}$, the tangent space $T_P Y_\delta$ does not contain P_0.*

Proof. Since Y_δ is irreducible and the locus of $P \in Y_\delta^{\mathrm{sm}}$ for which $T_P Y_\delta$ contains P_0 is Zariski closed, it suffices to find a point P such that $T_P Y_\delta$ does not contain P_0. Let $P \in Y_\delta^{\mathrm{sm}}$ be the point represented by the matrix whose (i,i)-entries are equal to 1 for $i = 2, \ldots, \delta$, and zero elsewhere. Let $\{e_1, \ldots, e_n\}$ denote the standard basis for K^n. Then

$$e_1 \in \ker P \quad \text{and} \quad \operatorname{im} P = \operatorname{span}(e_2, \ldots, e_\delta).$$

Since $P_0(e_1) = e_1$, we have

$$P_0(\ker P) \nsubseteq \operatorname{im} P,$$

and hence $P_0 \notin T_P Y_\delta$. This gives an example of P with the desired property. $\square$

Lemma 4.4. *If Y_δ is not a hypersurface, then for a general point $P \in Y_\delta^{\mathrm{sm}}$, we have*

$$\dim\left(T_{P_0} Y_\delta \cap T_P Y_\delta\right) \leq d - 2.$$

Proof. Since Y_δ is irreducible and the function $P \longmapsto \dim\left(T_{P_0} Y_\delta \cap T_P Y_\delta\right)$ is upper semicontinuous on Y_δ, it suffices to find a point P with

$$\dim\left(T_{P_0} Y_\delta \cap T_P Y_\delta\right) \leq d - 2,$$

or equivalently, such that $T_{P_0} Y_\delta \cap T_P Y_\delta$ has codimension at least 2 in $T_{P_0} Y_\delta$. Consider

$$P = \begin{pmatrix} 0 & 0 \\ 0 & I_{\delta-1} \end{pmatrix} \in Y_\delta^{\mathrm{sm}}.$$

Its tangent space $T_P Y_\delta$ consists of points represented by matrices of the form

$$\begin{pmatrix} 0 & *_{m-\delta+1,\, \delta-1} \\ *_{\delta-1,\, n-\delta+1} & *_{\delta-1,\, \delta-1} \end{pmatrix}.$$

Comparing this with (4.1), we see that the intersection $T_{P_0} Y_\delta \cap T_P Y_\delta$ consists of points represented by $m \times n$ matrices with $(m - \delta + 1) \times (n - \delta + 1)$ zero matrices as the top-left and bottom-right blocks.

- Suppose that

$$m-\delta+1 \le \delta-1 \qquad \text{or} \qquad n-\delta+1 \le \delta-1.$$

In this case, the two zero blocks do not overlap. Thus the linear subspace

$$T_{P_0}Y_\delta \cap T_P Y_\delta \subseteq T_{P_0}Y_\delta$$

is defined by $(m-\delta+1)(n-\delta+1)$ independent linear conditions. Hence

$$\operatorname{codim}_{T_{P_0}Y_\delta}(T_{P_0}Y_\delta \cap T_P Y_\delta) = (m-\delta+1)(n-\delta+1).$$

Note that this quantity equals 1 if and only if $m=n=\delta$.

- Suppose that

$$m-\delta+1 > \delta-1 \qquad \text{and} \qquad n-\delta+1 > \delta-1.$$

In this case, the two zero blocks overlap in a matrix of size

$$(m-2\delta+2) \times (n-2\delta+2).$$

Consequently, the linear subspace $T_{P_0}Y_\delta \cap T_P Y_\delta \subseteq T_{P_0}Y_\delta$ has codimension

$$\begin{aligned}&(m-\delta+1)(n-\delta+1)-(m-2\delta+2)(n-2\delta+2)\\ &=(\delta-1)(m+n-3\delta+3).\end{aligned}$$

This is never equal to 1. Indeed, if it were, then $\delta=2$ and $m+n=4$. However, the inequalities above imply $m+n>4(\delta-1)=4$, a contradiction.

In both cases, the intersection $T_{P_0}Y_\delta \cap T_P Y_\delta$ has codimension at least 2 in $T_{P_0}Y_\delta$ under the assumption that Y_δ is not a hypersurface. This completes the proof. □

4.2. **Finite linear sections with a tangency.** A linear subspace $L \subseteq \mathbb{P}^N$ is tangent at P_0 if and only if the intersection $L \cap T_{P_0}Y_\delta$ contains a line through P_0. The c-dimensional linear subspaces with this property form the variety

$$X_0 := \left\{ L \in \operatorname{Gr}(\mathbb{P}^c, \mathbb{P}^N) \;\middle|\; L \ni P_0,\ \dim(L \cap T_{P_0}Y_\delta) \ge 1 \right\}.$$

Our goal is to find an $L \in X_0$ that is tangent to Y_δ at P_0 with multiplicity two and transverse at all other intersection points. The proof proceeds in three steps:

(1) There exists $L \in X_0$ that is tangent to Y_δ at P_0 with multiplicity two.
(2) The locus of $L \in X_0$ that meet the singular locus $Y_{\delta-1} \subseteq Y_\delta$ has codimension ≥ 2.
(3) The locus of $L \in X_0$ that are tangent to Y_δ^{sm} at a second point has codimension ≥ 1.

Lemma 4.5. *The variety X_0 is irreducible with* $\dim X_0 = cd-1$.

Proof. The lines in $T_{P_0}Y_\delta$ passing through P_0 form the variety

$$\mathcal{B} := \left\{ \ell \in \operatorname{Gr}(\mathbb{P}^1, T_{P_0}Y_\delta) \;\middle|\; P_0 \in \ell \right\} \cong \mathbb{P}^{d-1}.$$

Consider the incidence correspondence

$$\begin{tikzcd} & I_0 := \{(L,\ell) \in X_0 \times \mathcal{B} \mid L \supseteq \ell\} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X_0 & & \mathcal{B}. \end{tikzcd}$$

For each $\ell \in \mathcal{B}$, the fiber $\pi_2^{-1}(\ell) \subseteq I_0$ consists of all pairs (L, ℓ) such that $L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N)$ contains ℓ. This gives I_0 the structure of a bundle over $\mathcal{B}$ with fiber $\mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2})$. It follows that I_0 is irreducible with

$$\dim I_0 = \dim \mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2}) + \dim \mathcal{B} = (c-1)d + (d-1) = cd - 1.$$

Moreover, a general $L \in X_0$ intersects $T_{P_0} Y_\delta$ in exactly one line. Hence π_1 is birational. It follows that X_0 is irreducible with $\dim X_0 = \dim I_0 = cd - 1$. $\square$

Lemma 4.6. *There exists $L \in X_0$ that is tangent to Y_δ at P_0 with multiplicity two.*

Proof. We work in the affine chart of $\mathbb{P}^N$ consisting of $m \times n$ matrices whose $(1,1)$-entry is equal to 1. Let $L \subseteq X_0$ be the linear subspace whose elements in this chart have the form

$$M := \begin{pmatrix} I_{\delta-1} & A \\ B & C \end{pmatrix}$$

where

- $A = (a_{ij})$ has $a_{ij} = 0$ for all $(i,j) \neq (1,1)$,
- $B = (b_{ij})$ has $b_{11} = a_{11}$ and $b_{ij} = 0$ for all $(i,j) \neq (1,1)$,
- $C = (c_{ij})$ has $c_{11} = 0$.

By the standard block row-reduction,

$$\mathrm{rank}(M) = \mathrm{rank} \begin{pmatrix} I_{\delta-1} & A \\ 0 & C - BA \end{pmatrix} = (\delta - 1) + \mathrm{rank}(C - BA).$$

Hence, $M \in L \cap Y_\delta$ if and only if $C = BA$, which holds if and only if

$$b_{11} a_{11} = a_{11}^2 = 0 \qquad \text{and} \qquad c_{ij} = 0 \quad \text{for all} \quad (i,j).$$

The equations $c_{ij} = 0$ define a line in L passing through P_0. On this line, $a_{11}^2 = 0$ defines a point of multiplicity two at P_0. Hence, L is a linear subspace with the desired properties. $\square$

Lemma 4.7. *The locus $X_s \subseteq X_0$ of linear subspaces $L \in X_0$ that meet the singular locus $Y_{\delta-1} \subseteq Y_\delta$ has dimension at most $cd - 3$.*

Proof. Consider the incidence correspondence

$$\begin{tikzcd} & I_s := \{(L, P) \in X_s \times Y_{\delta-1} \mid L \ni P\} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X_s & & Y_{\delta-1}. \end{tikzcd}$$

For each singular point $P \in Y_{\delta-1}$, every $L \in X_s$ containing P must also contain the line spanned by P_0 and P. Hence, if nonempty, the fiber $\pi_2^{-1}(P)$ can be identified with a subset of $\mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2})$. Therefore,

$$\begin{aligned} \dim I_s &\leq \dim Y_{\delta-1} + \dim \mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2}) \\ &= (N - (m - \delta + 2)(n - \delta + 2)) + (c-1)(N-c) \\ &= (d - (m + n - 2\delta + 3)) + (c-1)d \\ &= cd - (m + n - 2\delta + 3) \\ &\leq cd - 3. \end{aligned}$$

By the definition of X_s, the projection π_1 is surjective. Thus $\dim X_s \leq \dim I_s \leq cd - 3$. $\square$

4.3. **Finite linear sections with extra tangency.** Our next goal is to bound the dimension of the locus $X_t \subseteq X_0$ defined by

$$X_t := \left\{ L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N) \;\middle|\; L \text{ is tangent at } P_0 \text{ and some } P \in Y_\delta^{\mathrm{sm}} \setminus P_0 \right\}.$$

Lemma 4.8. *The locus $X_t' \subseteq X_t$ defined by*

$$X_t' := \{ L \in X_t \mid P \in T_{P_0} Y_\delta \}$$

has dimension at most $cd - 2$.

Proof. For each $P \in Y_\delta^{\mathrm{sm}} \setminus P_0$, the subspaces $L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N)$ containing both P_0 and P are precisely those that contain the line spanned by P_0 and P. Such subspaces are parametrized by $\mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2})$. By Lemma 4.2, the intersection $Y_\delta^{\mathrm{sm}} \cap T_{P_0} Y_\delta$ decomposes into a union of components, each of which is a cone with vertex P_0 and has dimension at most $d-1$. The cone structure implies that for any $P \in Y_\delta^{\mathrm{sm}} \cap T_{P_0} Y_\delta$ with $P \neq P_0$, all points on the line spanned by P_0 and P determine the same family of subspaces $L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N)$ containing both points. Consequently,

$$\begin{aligned} \dim X_t' &\leq \dim \mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2}) + (d-1) - 1 \\ &= (c-1)d + d - 2 \\ &= cd - 2. \end{aligned}$$

This yields the desired bound. □

It remains to analyze the open complement

$$X_t^\circ := X_t \setminus X_t' = \{ L \in X_t \mid P \notin T_{P_0} Y_\delta \}.$$

For each $L \in X_t$, there exist lines $\ell_0 \subseteq T_{P_0} Y_\delta$ and $\ell \subseteq T_P Y_\delta$ such that

$$P_0 \in \ell_0 \subseteq L \cap T_{P_0} Y_\delta \qquad \text{and} \qquad P \in \ell \subseteq L \cap T_P Y_\delta. \tag{4.2}$$

In view of this, we define

$$\begin{aligned} \mathcal{B}_0 &:= \left\{ \ell_0 \in \mathrm{Gr}(\mathbb{P}^1, T_{P_0} Y_\delta) \;\middle|\; P_0 \in \ell_0 \right\} \cong \mathbb{P}^{d-1}, \\ \mathcal{B}_1 &:= \left\{ \ell \ \in \mathrm{Gr}(\mathbb{P}^1, T_P\ Y_\delta) \;\middle|\; P\ \in \ell\ \right\} \cong \mathbb{P}^{d-1}, \end{aligned}$$

and consider the incidence correspondence

$$I_t \subseteq X_t^\circ \times Y_\delta^{\mathrm{sm}} \times \mathcal{B}_0 \times \mathcal{B}_1,$$

consisting of quadruples (L, P, ℓ_0, ℓ) satisfying (4.2). Each such quadruple satisfies:

- The lines ℓ_0 and ℓ are distinct.
- If ℓ_0 and ℓ intersect, the intersection point is different from P.

Indeed, if either of these conditions fails, then $P \in \ell_0 \subseteq T_{P_0} Y_\delta$, contradicting the assumption that $L \in X_t^\circ$. Hence I_t decomposes as a disjoint union

$$I_t = I_t' \cup I_t'',$$

where

$$\begin{aligned} I_t' &= \{ (L, P, \ell_0, \ell) \in I_t \mid \ell_0 \cap \ell = \emptyset \}, \\ I_t'' &= \{ (L, P, \ell_0, \ell) \in I_t \mid \ell_0 \cap \ell = \text{a point} \neq P \}. \end{aligned}$$

Lemma 4.9. $\dim I_t' \leq cd - 2$.

Proof. Consider the projection map

$$\pi' \colon I'_t \longrightarrow Y_\delta \times \mathcal{B}_0 \times \mathcal{B}_1 : (L, P, \ell_0, \ell) \longmapsto (P, \ell_0, \ell).$$

For each quadruple $(L, P, \ell_0, \ell) \in I'_t$, the condition $\ell_0 \cap \ell = \emptyset$ implies that their span

$$\Pi := \langle \ell_0, \ell \rangle \subseteq \mathbb{P}^N$$

is a 3-plane. Any L occurring in such a quadruple must contain Π. Hence, over a triple

$$(P, \ell_0, \ell) \in Y_\delta \times \mathcal{B}_0 \times \mathcal{B}_1,$$

the fiber $(\pi')^{-1}(P, \ell_0, \ell)$ is contained in $\mathrm{Gr}(\mathbb{P}^{c-4}, \mathbb{P}^{N-4})$. It follows that

$$\begin{aligned}
\dim I'_t &\leq \dim Y_\delta + \dim \mathcal{B}_0 + \dim \mathcal{B}_1 + \dim \mathrm{Gr}(\mathbb{P}^{c-4}, \mathbb{P}^{N-4}) \\
&= d + (d-1) + (d-1) + (c-3)(N-c) \\
&= 3d - 2 + (c-3)d \\
&= cd - 2.
\end{aligned}$$

This proves the desired upper bound. □

Lemma 4.10. $\dim I''_t \leq cd - 2$*, under the assumption that* Y_δ *is not a hypersurface.*

Proof. Assigning to each $(L, P, \ell_0, \ell) \in I''_t$ the intersection point $\ell_0 \cap \ell$ defines a morphism

$$\iota \colon I''_t \longrightarrow \mathbb{P}^N : (L, P, \ell_0, \ell) \longmapsto \ell_0 \cap \ell.$$

Let J_0 denote the fiber over P_0 and J_1 its complement. That is,

$$J_0 := \iota^{-1}(P_0), \qquad J_1 := I''_t \setminus J_0.$$

We first bound the dimension of J_0. Fix a point $P \in Y_\delta^{\mathrm{sm}}$ in the image of the projection

$$\pi \colon J_0 \longrightarrow Y_\delta^{\mathrm{sm}} : (L, P, \ell_0, \ell) \longmapsto P.$$

By the defining condition of I''_t, we have $P_0 = \ell_0 \cap \ell \neq P$. Thus, for any $(L, P, \ell_0, \ell) \in \pi^{-1}(P)$,

- ℓ is uniquely determined as the line through P_0 and P,
- ℓ_0 is a general line in $T_{P_0} Y_\delta$ passing through P_0, and
- L is a c-dimensional linear subspace containing the plane spanned by ℓ_0 and ℓ.

These imply that

$$\dim \pi^{-1}(P) \leq \dim \mathrm{Gr}(\mathbb{P}^{c-3}, \mathbb{P}^{N-3}) + (d-1) = (c-2)d + (d-1).$$

Moreover, we have $P_0 \in \ell \subseteq T_P Y_\delta$, so in particular $T_P Y_\delta$ contains P_0. By Lemma 4.3, the locus of such points $P \in Y_\delta^{\mathrm{sm}}$ has dimension at most $d - 1$. Consequently,

$$\dim J_0 \leq \dim \pi^{-1}(P) + (d-1) \leq (c-2)d + 2(d-1) = cd - 2.$$

Next, we bound the dimension of J_1. Consider the projection

$$\rho \colon J_1 \longrightarrow Y_\delta^{\mathrm{sm}} : (L, P, \ell_0, \ell) \longmapsto P.$$

For each $P \in Y_\delta^{\mathrm{sm}}$, the restriction of ι to the fiber $\rho^{-1}(P) \subseteq J_1$ defines a map

$$\iota_P := \iota|_{\rho^{-1}(P)} \colon \rho^{-1}(P) \longrightarrow T_{P_0} Y_\delta \cap T_P Y_\delta \cong \mathbb{P}^e.$$

By the defining conditions of I''_t and J_1, we have

$$P_0 \neq \ell_0 \cap \ell \neq P.$$

Thus, for each fiber $(L, P, \ell_0, \ell) \in \rho^{-1}(Q)$,

- ℓ_0 is uniquely determined as the line through P_0 and Q,
- ℓ is uniquely determined as the line through P and Q, and
- L is a c-dimensional linear subspace containing the plane spanned by ℓ_0 and ℓ.

Moreover, since Y_δ is not a hypersurface, Lemma 4.4 implies $e \leq d-2$. Hence,

$$\dim \rho^{-1}(P) \leq \dim \operatorname{Gr}(\mathbb{P}^{c-3}, \mathbb{P}^{N-3}) + e \leq (c-2)d + (d-2).$$

Consequently,

$$\dim J_1 \leq \dim \rho^{-1}(P) + d \leq (c-2)d + (d-2) + d = cd - 2.$$

It follows that

$$\dim I_t'' \leq \max\{\dim J_0, \dim J_1\} \leq cd - 2,$$

which proves the desired bound. □

Lemma 4.11. $\dim X_t \leq cd - 2$, *under the assumption that Y_δ is not a hypersurface.*

Proof. Due to Lemma 4.8, it remains to prove that

$$\dim X_t^\circ \leq cd - 2.$$

Each $L \in X_t^\circ$ is tangent at both P_0 and some point $P \in Y_\delta^{\mathrm{sm}} \setminus P_0$. Hence there exist lines

$$\ell_0 \subseteq L \cap T_{P_0} Y_\delta \qquad \text{and} \qquad \ell \subseteq L \cap T_P Y_\delta$$

passing through P_0 and P, respectively. This shows that the projection map

$$I_t \longrightarrow X_t^\circ : (L, P, \ell_0, \ell) \longmapsto L$$

is surjective. It follows that

$$\dim X_t^\circ \leq \dim I_t \leq \max\{\dim I_t', \dim I_t''\},$$

which is at most $cd - 2$ by Lemmas 4.9 and 4.10. □

Theorem 4.12. *Every determinantal variety $Y_\delta \subseteq \mathbb{P}^N$ is quasireflexive. That is, there exists a linear subspace $L \subseteq \mathbb{P}^N$ of dimension $N - \dim Y_\delta$ that intersects Y_δ in $\deg(Y_\delta) - 1$ points at which Y_δ is smooth.*

Proof. The hypersurface case is proved in Proposition 4.1. For non-hypersurfaces, Lemma 4.6 implies the existence of a Zariski open subset $U \subseteq X_0$ parametrizing linear subspaces whose tangency with Y_δ at P_0 has multiplicity two. By Lemmas 4.5, 4.7 and 4.11, the locus in X_0 parametrizing linear subspaces that meet Y_δ non-transversely at points other than P_0 form a proper closed subset. Therefore, U contains a member that is tangent to Y_δ at P_0 with multiplicity two and intersects Y_δ transversely at all other points. □

Appendix A. Reflexivity of determinantal varieties

For curves over a field of characteristic not equal to 2, reflexivity and quasireflexivity are equivalent notions by [Ent21, Proposition 2.1]. In higher dimensions, following the second paragraph of the proof of [BH86, Theorem at p. 906], one deduces that:

Proposition A.1. *Every reflexive variety over a field of characteristic not equal to 2 is quasireflexive.*

In what follows, we prove that determinantal varieties defined over a field of arbitrary characteristic are reflexive. By Proposition A.1, this already implies their quasireflexivity in characteristic different from 2. However, it does not address the case of characteristic 2, which is of particular importance in coding theory.

We begin by recalling the definition of reflexivity. Let $Y \subseteq \mathbb{P}^N$ be a projective variety defined over an algebraically closed field K of arbitrary characteristic. Its *conormal variety*

$$C(Y) \subseteq \mathbb{P}^N \times \left(\mathbb{P}^N\right)^*$$

is the Zariski closure of the set of pairs $(P, H) \in \mathbb{P}^N \times \left(\mathbb{P}^N\right)^*$ where

- P belongs to the smooth locus $Y^{\mathrm{sm}} \subseteq Y$, and
- H is a hyperplane in $\mathbb{P}^N$ containing the tangent space $T_P Y \subseteq \mathbb{P}^N$.

Consider the two projection maps

$$\begin{tikzcd} & C(Y) \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ Y & & \left(\mathbb{P}^N\right)^* . \end{tikzcd}$$

The *dual variety* of Y is defined as the image of the second projection

$$Y^* := \pi_2(C(Y)) \subseteq \left(\mathbb{P}^N\right)^*,$$

and Y is *reflexive* if $C(Y) \cong C(Y^*)$ under the natural isomorphisms

$$\mathbb{P}^N \times \left(\mathbb{P}^N\right)^* \cong \left(\mathbb{P}^N\right)^* \times \mathbb{P}^N \cong \left(\mathbb{P}^N\right)^* \times \left(\mathbb{P}^N\right)^{**}. \tag{A.1}$$

Let $Y_\delta \subseteq \mathbb{P}^N = \mathbb{P}^{mn-1}$ be the determinantal variety over K consisting of $m \times n$ matrices, where $m \geq n$, of rank $< \delta$. Recall that the tangent space at a smooth point $P \in Y_\delta^{\mathrm{sm}}$ is given by

$$T_P Y_\delta = \left\{ Q \in \mathbb{P}^N \;\middle|\; Q(\ker P) \subseteq \operatorname{im} P \right\}.$$

On the other hand, there is a nondegenerate bilinear form

$$\langle -, - \rangle \colon K^{mn} \times K^{mn} \longrightarrow K, \quad (A, B) \longmapsto \operatorname{tr}(AB^T).$$

In coordinates, if we write $A = (a_{ij})$ and $B = (b_{ij})$, then

$$\langle A, B \rangle = \sum_{i,j} a_{ij} b_{ij}.$$

We identify $(\mathbb{P}^N)^*$ with $\mathbb{P}^N$ via the induced isomorphism, which is explicitly given by

$$\mathbb{P}^N \xrightarrow{\sim} (\mathbb{P}^N)^* : P \longmapsto H_P = \left\{ \textstyle\sum_{i,j} p_{ij} x_{ij} = 0 \right\}$$

where (p_{ij}) is any matrix representing P.

Lemma A.2. *Let $I_{\delta-1}$ denote the $(\delta-1) \times (\delta-1)$ identity matrix. The fiber over*

$$P_0 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & 0 \end{pmatrix} \in Y_\delta$$

under the projection $\pi_1 \colon C(Y_\delta) \longrightarrow Y_\delta$ has the form

$$\pi_1^{-1}(P_0) = \left\{ \left(P_0, \begin{pmatrix} 0 & 0 \\ 0 & *_{m-\delta+1,\, n-\delta+1} \end{pmatrix} \right) \right\}$$

where $*_{\alpha,\beta}$ *denotes an arbitrary* $\alpha \times \beta$ *matrix. In particular,*

$$\pi_2\left(\pi_1^{-1}(P_0)\right) \subseteq Y_{n-\delta+2}.$$

Proof. Every $Q \in T_{P_0}Y_\delta$ is represented by a matrix of the form

$$\begin{pmatrix} *_{\delta-1,\,\delta-1} & *_{\delta-1,\,n-\delta+1} \\ *_{m-\delta+1,\,\delta-1} & 0 \end{pmatrix}.$$

Hence, H_P contains $T_{P_0}Y_\delta$ if and only if P is represented by

$$\widetilde{P} = \begin{pmatrix} 0 & 0 \\ 0 & *_{m-\delta+1,\,n-\delta+1} \end{pmatrix}.$$

Since $(P_0, P) \in \mathbb{P}^N \times (\mathbb{P}^N)^*$ belongs to $\pi_1^{-1}(P_0)$ if and only if $T_{P_0}Y_\delta \subseteq H_P$, this proves the first statement. The second statement follows immediately from the expression of $\widetilde{P}$. ☐

Lemma A.3. *For any* $A \in \mathrm{GL}_m(K)$ *and* $B \in \mathrm{GL}_n(K)$*,*

$$T_P Y_\delta \subseteq H_Q \iff T_{APB}Y_\delta \subseteq H_{A^\dagger Q B^\dagger}$$

where $A^\dagger = (A^{-1})^T$ *and* $B^\dagger = (B^{-1})^T$*.*

Proof. Note that

$$\mathrm{tr}\left(P\,Q^T\right) = \mathrm{tr}\left((APB)\,(A^\dagger Q B^\dagger)^T\right). \tag{A.2}$$

Using the identities

- $\ker(APB) = \ker(PB) = B^{-1}\ker P$,
- $\mathrm{im}(APB) = \mathrm{im}(AP) = A\ \mathrm{im}\,P$,

we deduce for any $M \in \mathrm{Hom}(K^n, K^m)$ that

$$\begin{aligned}(AMB)\ \ker(APB) \subseteq \mathrm{im}(APB) &\iff (AMB)\,(B^{-1}\ker P) \subseteq A\ \mathrm{im}\,P \\ &\iff M\ \ker P \subseteq \mathrm{im}\,P.\end{aligned}$$

It follows that

$$A\,(T_P Y_\delta)\,B = T_{APB}Y_\delta. \tag{A.3}$$

Consequently,

$$\begin{aligned} T_P Y_\delta \subseteq H_Q &\iff \langle T_P Y_\delta,\, Q\rangle = 0 \\ \text{(by (A.2))} \quad &\iff \left\langle A\,(T_P Y_\delta)\,B,\ A^\dagger Q B^\dagger\right\rangle = 0 \\ \text{(by (A.3))} \quad &\iff \left\langle T_{APB}Y_\delta,\ A^\dagger Q B^\dagger\right\rangle = 0 \iff T_{APB}Y_\delta \subseteq H_{A^\dagger Q B^\dagger}.\end{aligned}$$

This completes the proof. ☐

Lemma A.4. *For any* $(P, Q) \in Y_\delta^{\mathrm{sm}} \times Y_{n-\delta+2}^{\mathrm{sm}}$*, the inclusion* $T_P Y_\delta \subseteq H_Q$ *holds if and only if there exist matrices* $A \in \mathrm{GL}_m(K)$ *and* $B \in \mathrm{GL}_n(K)$ *such that*

$$APB = P_0, \qquad A^\dagger Q B^\dagger = \begin{pmatrix} 0 & 0 \\ 0 & I_{n-\delta+1} \end{pmatrix},$$

where $A^\dagger = (A^{-1})^T$ *and* $B^\dagger = (B^{-1})^T$*.*

Proof. If there exist matrices A and B with the given identities, then

$$T_{APB}Y_\delta = T_{P_0}Y_\delta \subseteq H_{A^\dagger QB^\dagger},$$

which implies $T_PY_\delta \subseteq H_Q$ by Lemma A.3.

It remains to prove the converse. Since P has rank $\delta - 1$, there exist $A_1 \in \mathrm{GL}_m(K)$ and $B_1 \in \mathrm{GL}_n(K)$ such that $A_1PB_1 = P_0$. By Lemma A.3,

$$T_PY_\delta \subseteq H_Q \iff T_{A_1PB_1}Y_\delta = T_{P_0}Y_\delta \subseteq H_{A_1^\dagger QB_1^\dagger}$$

The last inclusion relation $T_{P_0}Y_\delta \subseteq H_{A_1^\dagger QB_1^\dagger}$, together with Lemma A.2, implies

$$A_1^\dagger QB_1^\dagger = \begin{pmatrix} 0 & 0 \\ 0 & C \end{pmatrix}$$

for some $(m-\delta+1) \times (n-\delta+1)$ matrix C. Since $\operatorname{rank} C = \operatorname{rank} Q = n-\delta+1$, there exist $A' \in \mathrm{GL}_{m-\delta+1}(K)$ and $B' \in \mathrm{GL}_{n-\delta+1}(K)$ such that

$$A'^\dagger CB'^\dagger = \begin{pmatrix} 0 \\ I_{n-\delta+1} \end{pmatrix}_{(m-\delta+1)\times(n-\delta+1)}$$

Consider now the matrices

$$A_2 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & A' \end{pmatrix} \in \mathrm{GL}_m(K) \quad \text{and} \quad B_2 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & B' \end{pmatrix} \in \mathrm{GL}_n(K).$$

Define $A = A_2A_1$ and $B = B_1B_2$. Then they satisfy the desired property:

$$APB = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & 0 \end{pmatrix}, \qquad A^\dagger QB^\dagger = \begin{pmatrix} 0 & 0 \\ 0 & I_{n-\delta+1} \end{pmatrix}.$$

This completes the proof. $\square$

Proposition A.5. *For any $(P,Q) \in Y_\delta^{\mathrm{sm}} \times Y_{n-\delta+2}^{\mathrm{sm}}$,*

$$T_PY_\delta \subseteq H_Q \iff T_QY_{n-\delta+2} \subseteq H_P.$$

In particular, Y_δ has dual variety $Y_{n-\delta+2}$ and is reflexive.

Proof. The statement of Lemma A.4 is symmetric in P and Q up to permutations of rows and columns, since $n-(n-\delta+2)+2 = \delta$. Therefore, the first statement holds. This implies that Y_δ and $Y_{n-\delta+2}$ are dual to each other, and that there is an isomorphism

$$\left\{(P,Q) \in Y_\delta^{\mathrm{sm}} \times Y_{n-\delta+2}^{\mathrm{sm}} \;\middle|\; T_PY_\delta \subseteq H_Q\right\} \cong \left\{(Q,P) \in Y_{n-\delta+2}^{\mathrm{sm}} \times Y_\delta^{\mathrm{sm}} \;\middle|\; T_QY_{n-\delta+2} \subseteq H_P\right\}.$$

Taking the Zariski closures of both sides yields an isomorphism

$$C(Y_\delta) \cong C(Y_{n-\delta+2}),$$

which is compatible with isomorphism (A.1). This shows that Y_δ is reflexive. $\square$

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